Two-minute cadence for a pre-registered transit of the 36.3-day sub-Neptune candidate around TIC 402710280

Rishit Vora (Independent Researcher | rishitv@protonmail.com)

1 DDT Justification

TIC 402710280 (TYC 9041-03677-1; $T = 9.24$, $V = 9.98$) hosts a transiting sub-Neptune candidate with $P = 36.33986 \pm 0.00051$ d, designated TIC 402710280.01 in Vora (2026a). *TESS* has observed the star only in full-frame images (FFIs), and no TOI, CTOI or SPOC threshold-crossing event exists for it. The period was fixed only by the transit of 2026 March 29 in Sector 102, consecutive with that of Sector 100 and after the Cycle 9 Guest Investigator deadline of 2026 March 10. It is on none of the Cycle 9 two-minute or 20-s target lists for Sectors 113 and 114 (ExoFOP, MAST and the target lists checked 2026 October 18). According to `tess-point` 0.9.5 (Burke et al., 2020), with provisional pointings, *TESS* will observe it in those two sectors and not again through Sector 134 (2028 September). The transit of 2027 February 19, in Sector 114, is therefore the only one *TESS* will record before late 2028. Without a DDT allocation it will be recorded in 200-s FFIs only, with no target pixel files and no SPOC light curve.

2 Scientific Justification

Four transits (Sectors 39, 93, 100 and 102; Figure 1), measured in three independent reductions (TESS-SPOC, QLP and our own FFI photometry), have consistent depths ($\chi^2 = 2.5$ for 3 degrees of freedom; weighted mean 2092 ± 47 ppm) and a duration of 5.27 ± 0.05 h. Every shorter alias predicts transits in covered data that are absent: $P/2$ in Sectors 39, 101 and 102 at $\geq 28.5\sigma$, and $P/3$ to $P/5$ at $\geq 18.6\sigma$. For the TIC stellar parameters ($T_{\rm eff} = 5182$ K, $R_\star = 0.82\,R_\odot$, $M_\star = 0.88\,M_\odot$), the candidate has $R_p = 3.75 \pm 0.22\,R_\oplus$, $a = 0.206$ AU and $T_{\rm eq} \approx 500$ K. Difference-image centroids place the signal $1.5'' \pm 1.3''$ from the target. The only *Gaia* source near enough to contribute, $3.5''$ away and 354 times fainter, would need a 74% eclipse, whose shape is disfavoured for any main-sequence pair ($\Delta\chi^2 \geq 142$). No secondary eclipse is detected. `TRICERATOPS` (Giacalone et al., 2021) gives FPP $= 0.003 \pm 0.004$ and NFPP $< 10^{-6}$; without high-resolution imaging we treat these as supporting evidence, not as validation. A mass of about 14 $M_\oplus$ would give a radial-velocity semi-amplitude of about 2.8 m s^{-1}.

Two-minute data would serve three purposes. The 36-day transit falls in Sector 114 only; Sector 113 supplies the out-of-transit baseline in the same aperture and half of the 3.78-day events below.

(1) The pre-registered test. The transit times scatter about a linear ephemeris by 21.8 min rms, against formal uncertainties of 1.7–3.3 min. We find no instrumental cause, and four epochs cannot separate timing systematics from transit-timing variations. On 2026 October 15 we registered the epoch-57 prediction ($\mathrm{BJD_{TDB}}$ 2461455.9785 ± 30 min), the data to be used, and the criteria that classify the outcome (Vora, 2026b). The test was designed around SPOC two-minute light curves, and its check of the signal's position requires target pixel files.

(2) An independent mid-time. The largest contribution to the scatter is the interval between

the Sector 100 and 102 transits, both from our own photometry, which exceeds P by 29 min. A SPOC mid-time at epoch 57, independent of our reductions, tests whether that interval reflects our photometry or the orbit.

(3) The 3.78-day signal. The light curve also contains a 407 ± 26 ppm, 1.3-h signal with $P = 3.78371$ d, whose source, the target (as a near-grazing transit) or the 3.5″ neighbour, the present data cannot identify (Vora, 2026a). Two-minute cadence would resolve the ingress of the dozen events in the two sectors, which tests the grazing geometry, and the target pixel files would tighten the difference-image centroid, now $5.2'' \pm 6.7''$ from the target.

3 Target Justification and Feasibility

We request two-minute cadence for one target, TIC 402710280 (RA 248.5236°, Dec −63.5497°), in Sectors 113 and 114, at priority 1.0. `tess-point` places it on camera 4, CCD 4 (column 385, row 209) in Sector 113 and on camera 4, CCD 1 (column 646, row 475) in Sector 114. The target is fainter than 7th magnitude and needs no enlarged pixel stamp.

The epoch-57 transit is predicted for 2027 February 19 at 11:29 UTC with $\sigma = 30$ min (34 min when the covariance of T_0 and P is included), six days after the start of Sector 114 and well inside its first orbit. The neighbouring transits, on 2027 January 14 and March 27, fall outside both sectors.

In FFIs each transit depth is measured to 84–144 ppm, so a single transit is detected at 15–25σ; SPOC two-minute photometry should match this. Cadence barely changes the timing precision, which is set by photon noise; the gains are SPOC photometry with pipeline systematics correction, target pixel files that allow apertures to be chosen against the 3.5″ neighbour, and ingress and egress (about 14 min each) sampled by seven cadences instead of four. The ingress shape is what distinguishes a planet from a diluted eclipse of the neighbour.

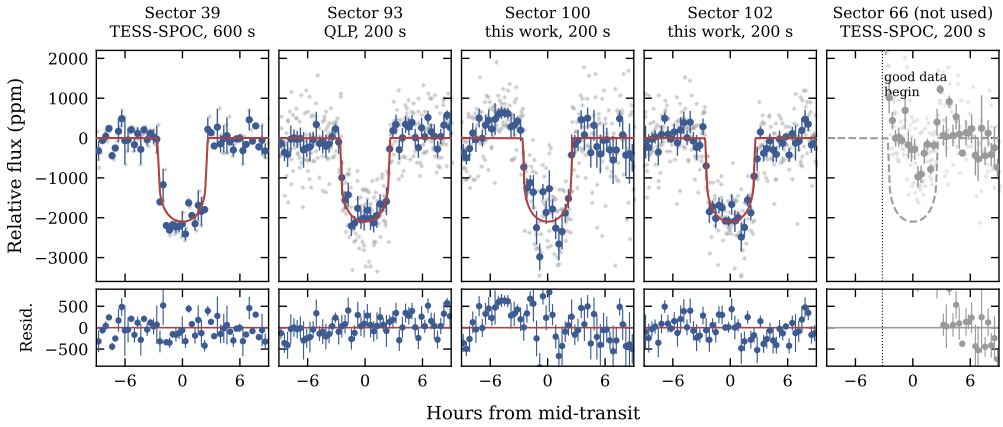


Figure 1: The four transits (20-min bins) with the median model and residuals; the Sector 66 window (right), lost to scattered light, is excluded from all fits (Vora, 2026a).

References

Burke, C. J., Levine, A., Fausnaugh, M., et al. 2020, `tess-point`, ascl:2003.001

Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, AJ, 161, 24

Vora, R. 2026a, preprint, doi:10.5281/zenodo.23390317; data and code, doi:10.5281/zenodo.23390274

Vora, R. 2026b, pre-registered test of the epoch-57 and 59 transits, doi:10.5281/zenodo.23390325